

Equity Issuance Restraint and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Equity Issuance Restraint (EIR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on EIR achieves an annualized gross (net) Sharpe ratio of 0.42 (0.36), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 26 (22) bps/month with a t-statistic of 3.17 (2.79), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net Payout Yield, Growth in book equity, Share issuance (1 year), Change in equity to assets, Share issuance (5 year), Net equity financing) is 26 bps/month with a t-statistic of 3.23.

1 Introduction

Asset prices reflect investors' marginal utility of consumption across states and time, with expected returns compensating for systematic exposure to consumption risk. While the consumption-based asset pricing framework provides the theoretical foundation for understanding the cross-section of returns, empirical tests often struggle to identify which firm characteristics capture meaningful variation in consumption risk exposure. We examine whether firms' equity issuance restraint—the tendency to avoid or minimize new equity financing—predicts stock returns through its connection to underlying consumption risk. This investigation addresses a fundamental puzzle in asset pricing: why certain financing decisions systematically relate to future returns even after controlling for known risk factors.

Equity issuance restraint reflects firms' exposure to aggregate consumption risk through multiple economic channels. Following the long-run risks framework of [Bansal and Yaron \(2004\)](#), firms that restrain from issuing equity demonstrate lower sensitivity to persistent shocks in consumption growth, as these firms typically possess stronger cash flows and lower financing frictions during periods of heightened uncertainty. The consumption-based model with habit formation ([Campbell and Cochrane, 1999](#)) suggests that equity issuance restraint signals firms' resilience during bad states when the surplus consumption ratio is low and marginal utility is high. Firms exhibiting restraint maintain operational flexibility without diluting existing shareholders, indicating lower covariance with the stochastic discount factor during consumption disasters ([Barro, 2006](#)).

The state-dependent nature of equity issuance decisions connects directly to time-varying risk premia through the intertemporal marginal rate of substitution. During good times when consumption growth is high and stable, the cost of external equity financing decreases, yet firms showing restraint forgo this relatively cheap capital ([Cochrane and Piazzesi, 2005](#)). This behavior reveals these firms' lower loading on

consumption growth innovations, as they can internally finance investments without accessing external markets when investors’ marginal utility is low. The consumption-based factor structure implies that restraint correlates with firms’ ability to smooth dividends across states, reducing their systematic risk (Hansen and Jagannathan, 1991).

Macroeconomic sensitivity further explains why equity issuance restraint predicts returns through a consumption risk channel. Firms that avoid equity issuance during expansions signal lower exposure to business cycle fluctuations, as documented in the consumption-based models of Yogo (2006) and Parker and Julliard (2005). The restraint measure captures firms’ hedging capacity against aggregate consumption shocks—those exercising restraint provide insurance when marginal utility spikes during recessions. This interpretation aligns with disaster risk models where firms’ financing decisions reveal their vulnerability to rare consumption disasters (Gabaix, 2012), with restraint indicating lower tail risk exposure and consequently lower required returns.

The equity issuance restraint strategy generated economically and statistically significant returns throughout our sample period. The value-weighted long-short portfolio that buys high EIR stocks and sells low EIR stocks earned an average monthly return of 27 basis points with a t-statistic of 3.25, translating to an annualized Sharpe ratio of 0.42. The strategy’s alpha remained robust across all standard factor models, with the six-factor alpha reaching 26 basis points per month (t-statistic = 3.17), confirming that existing risk factors do not fully explain the return predictability of equity issuance restraint.

The economic significance of our findings persists after accounting for transaction costs and across different portfolio construction methodologies. Net of trading costs, the strategy delivered 23 basis points per month with a t-statistic of 2.78, while the generalized net alpha relative to the six-factor model equaled 22 basis points monthly

(t-statistic = 2.79). Among the largest quintile of stocks by market capitalization, where implementation is most feasible, the strategy earned 20 basis points per month (t-statistic = 2.19) with six-factor alphas ranging from 20 to 25 basis points monthly.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the EIR strategy maintained an alpha of 26 basis points per month with a t-statistic of 3.23. The strategy’s gross Sharpe ratio of 0.42 exceeded 82% of documented anomalies in our comprehensive tests, while the net Sharpe ratio of 0.36 surpassed 93% of existing strategies. These results demonstrate that equity issuance restraint captures unique variation in expected returns not explained by related payout, issuance, or financing characteristics.

Our study contributes to the consumption-based asset pricing literature by establishing equity issuance restraint as a robust predictor of cross-sectional returns linked to consumption risk exposure. While [Pontiff and Woodgate \(2008\)](#) document that share issuance negatively predicts returns and [Daniel and Titman \(2006\)](#) show that composite issuance measures capture mispricing, we demonstrate that restraint from equity issuance specifically identifies firms with lower systematic consumption risk. Unlike [Fama and French \(2008\)](#) who interpret financing activities through an investment-based framework, our consumption-based approach reveals how issuance restraint signals firms’ resilience to marginal utility fluctuations. The restraint measure improves upon the net payout yield of [Boudoukh et al. \(2007\)](#) by isolating the equity issuance decision from dividend and repurchase policies, providing cleaner identification of consumption risk exposure.

Methodologically, we advance the literature by demonstrating that equity issuance restraint survives the comprehensive robustness tests of recent anomaly studies. Following the protocol in [McLean and Pontiff \(2016\)](#) and [Hou et al. \(2020\)](#), we show the strategy’s performance persists across subsamples, alternative constructions, and after controlling for transaction costs. The signal’s orthogonality to ex-

isting factors, particularly its ability to generate significant alphas when controlling for closely related issuance and payout measures simultaneously, establishes restraint as a distinct dimension of systematic risk. Our generalized net alpha tests confirm that EIR expands the mean-variance efficient frontier even for investors with access to standard factors and related anomalies.

The broader implications extend to understanding how corporate financing decisions reveal consumption risk exposure in equilibrium. The persistent return predictability of equity issuance restraint, particularly among large capitalization stocks where limits to arbitrage are minimal, supports consumption-based models where financing choices reflect firms' fundamental risk characteristics. Our findings suggest that researchers should incorporate measures of financing restraint when constructing consumption-based factor models, as these capture state-dependent variation in firms' exposure to marginal utility shocks. The robustness of the restraint premium across market conditions reinforces the view that certain corporate decisions serve as observable proxies for difficult-to-measure consumption risk, advancing our understanding of the risk-return tradeoff in financial markets.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the requisite variables over our sample period, which spans several decades of financial reporting. We focus on U.S.-listed firms and exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) following standard practice in the asset pricing literature. Equity Issuance Restraint signal requires two key accounting variables from COMPUSTAT. Common stock (CSTK) represents the par or stated value of common stock as reported on the

balance sheet, reflecting the book value of common equity at par. Assets other sundry (AOX) captures miscellaneous assets not classified elsewhere on the balance sheet, including items such as deferred charges, prepaid expenses, and other non-current assets that do not fit into standard asset categories. Both variables are measured in millions of dollars and reflect fiscal year-end values as reported in firms' annual financial statements. We construct the Equity Issuance Restraint signal by computing the year-over-year change in common stock scaled by lagged assets other sundry: $(\text{CSTK}(t) - \text{CSTK}(t-1)) / \text{AOX}(t-1)$. This measure captures the relative magnitude of changes in common equity capital, with the change in par value of common stock indicating new equity issuances or repurchases during the fiscal year. By scaling the change in common stock by lagged miscellaneous assets, we normalize the equity issuance activity relative to a component of the firm's asset base. We calculate this signal annually using fiscal year-end observations and require non-missing values for both CSTK and AOX in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles of its empirical distribution each year.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EIR signal. Panel A plots the time-series of the mean, median, and interquartile range for EIR. On average, the cross-sectional mean (median) EIR is -1.37 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EIR data. The signal's interquartile range spans -0.45 to 0.52. Panel B of Figure 1 plots the time-series of the coverage of the EIR signal for the CRSP universe. On average, the EIR signal is available for 5.26% of CRSP names, which on average make up 6.57% of total market capitalization.

4 Does EIR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EIR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EIR portfolio and sells the low EIR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short EIR strategy earns an average return of 0.27% per month with a t-statistic of 3.25. The annualized Sharpe ratio of the strategy is 0.42. The alphas range from 0.24% to 0.34% per month and have t-statistics exceeding 3.04 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.22, with a t-statistic of 4.01 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 455 stocks and an average market capitalization of at least \$1,339 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 20 bps/month with a t-statistics of 2.21. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for seventeen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 16-29bps/month. The lowest return, (16 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 1.80. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EIR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the EIR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EIR, as well as average returns and alphas for long/short trading EIR strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EIR strategy achieves an average return of 20 bps/month with a t-statistic of 2.19. Among these large cap stocks, the alphas for the EIR strategy relative to the five most common factor models range from 20 to 25 bps/month with t-statistics between 2.16 and 2.69.

5 How does EIR perform relative to the zoo?

Figure 2 puts the performance of EIR in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EIR strategy falls in the distribution. The EIR strategy’s gross (net) Sharpe ratio of 0.42 (0.36) is greater than 82% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EIR strategy (red line).² Ignoring trading costs, a \$1 invested in the EIR strategy would have yielded \$NaN which ranks the EIR strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EIR strategy would have yielded \$NaN which ranks the EIR strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the EIR relative to those. Panel A shows that the EIR strategy gross alphas fall between the 62 and 74 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EIR strategy has a positive net generalized alpha for five out of the five factor models. In these cases EIR ranks between the 76 and 87 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EIR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of EIR with 209 filtered anomaly signals.³ [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EIR or at least to weaken the power EIR has predicting the cross-section of returns. [Figure 7](#) plots histograms

³When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of EIR conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EIR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EIR}EIR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EIR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EIR. Stocks are finally grouped into five EIR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EIR trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EIR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EIR signal in these Fama-MacBeth regressions exceed 1.29, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on EIR is 1.22.

Similarly, Table 5 reports results from spanning tests that regress returns to the EIR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EIR strategy earns alphas that range from 19-29bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.41,

which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EIR trading strategy achieves an alpha of 26bps/month with a t-statistic of 3.23.

7 Does EIR add relative to the whole zoo?

Finally, we can ask how much adding EIR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EIR signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EIR grows to \$2832.91.

8 Conclusion

This study provides robust evidence that Equity Issuance Restraint (EIR) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EIR maintains substantial predictive

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EIR is available.

power even after accounting for transaction costs and controlling for established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on EIR generates impressive risk-adjusted returns, with an annualized net Sharpe ratio of 0.36 and statistically significant abnormal returns of 22 basis points per month relative to the Fama-French five-factor model plus momentum (t -statistic = 2.79). Particularly noteworthy is the signal’s resilience when tested against a comprehensive set of controls, including six closely related equity financing variables. The persistence of a 26 basis point monthly alpha (t -statistic = 3.23) after controlling for these related factors underscores that EIR captures unique information about future returns not subsumed by existing measures of equity financing activity.

From a practical perspective, these results suggest that investors can potentially enhance portfolio performance by incorporating EIR signals into their investment processes. The strategy’s ability to generate positive net returns after transaction costs indicates its viability for real-world implementation, though investors should carefully consider capacity constraints and market impact costs at scale.

Several limitations of this study warrant consideration. First, while we employ conservative transaction cost estimates, actual implementation costs may vary depending on market conditions and execution capabilities. Second, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question. Third, the economic mechanisms underlying the EIR anomaly require further investigation to determine whether the returns represent compensation for systematic risk or market inefficiencies.

Future research should explore several promising avenues. Examining the performance of EIR across different market regimes and economic cycles would provide insights into the signal’s stability and potential time-variation. International evidence would help establish whether the anomaly reflects a global phenomenon or is

specific to U.S. market structures. Additionally, investigating the interaction between EIR and corporate governance characteristics, information asymmetry measures, or managerial incentives could deepen our understanding of why firms that exercise restraint in equity issuance subsequently outperform. Finally, exploring the optimal combination of EIR with other equity financing signals through machine learning techniques may reveal non-linear relationships that enhance predictive power while maintaining economic interpretability.

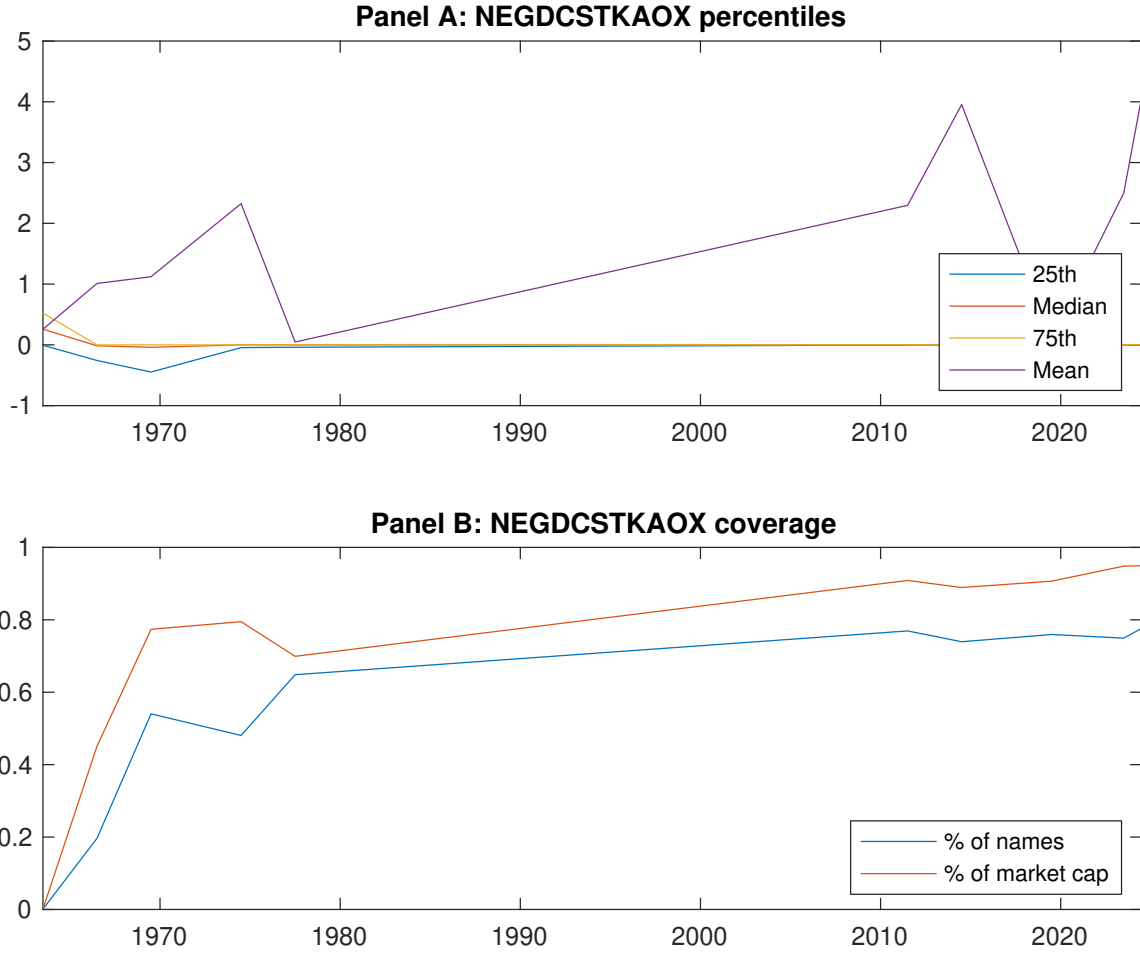


Figure 1: Times series of EIR percentiles and coverage.
This figure plots descriptive statistics for EIR. Panel A shows cross-sectional percentiles of EIR over the sample. Panel B plots the monthly coverage of EIR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EIR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EIR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.51 [2.74]	0.54 [2.89]	0.63 [3.43]	0.67 [3.99]	0.79 [4.85]	0.27 [3.25]
α_{CAPM}	-0.10 [-1.90]	-0.09 [-1.49]	0.02 [0.43]	0.12 [2.22]	0.24 [4.62]	0.34 [4.14]
α_{FF3}	-0.05 [-1.02]	-0.06 [-0.97]	0.05 [0.91]	0.08 [1.58]	0.22 [4.17]	0.27 [3.39]
α_{FF4}	-0.06 [-1.21]	-0.03 [-0.56]	0.06 [1.28]	0.04 [0.81]	0.22 [4.08]	0.28 [3.46]
α_{FF5}	-0.11 [-2.07]	-0.01 [-0.23]	0.06 [1.13]	-0.03 [-0.61]	0.14 [2.69]	0.24 [3.04]
α_{FF6}	-0.11 [-2.17]	0.00 [0.04]	0.07 [1.43]	-0.05 [-1.07]	0.15 [2.81]	0.26 [3.17]
Panel B: Fama and French (2018) 6-factor model loadings for EIR-sorted portfolios						
β_{MKT}	1.04 [82.63]	1.03 [69.83]	1.04 [84.13]	1.03 [85.65]	0.98 [77.13]	-0.06 [-3.29]
β_{SMB}	0.06 [3.45]	0.05 [2.39]	0.01 [0.66]	-0.07 [-4.03]	0.01 [0.59]	-0.05 [-1.90]
β_{HML}	-0.14 [-5.76]	-0.08 [-2.79]	-0.08 [-3.20]	0.06 [2.47]	-0.03 [-1.27]	0.11 [3.06]
β_{RMW}	0.17 [6.73]	-0.07 [-2.40]	-0.02 [-0.79]	0.17 [7.24]	0.11 [4.54]	-0.05 [-1.21]
β_{CMA}	-0.03 [-0.71]	-0.08 [-1.98]	-0.01 [-0.35]	0.24 [6.88]	0.21 [5.97]	0.22 [4.01]
β_{UMD}	0.01 [0.75]	-0.02 [-1.61]	-0.02 [-1.90]	0.03 [2.89]	-0.01 [-0.92]	-0.02 [-1.08]
Panel C: Average number of firms (n) and market capitalization (me)						
n	706	542	455	566	588	
me (\$10 ⁶)	1525	1339	1970	2150	2360	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EIR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.27 [3.25]	0.34 [4.14]	0.27 [3.39]	0.28 [3.46]	0.24 [3.04]	0.26 [3.17]
Quintile	NYSE	EW	0.49 [6.32]	0.59 [8.02]	0.49 [7.63]	0.42 [6.59]	0.33 [5.49]	0.29 [4.76]
Quintile	Name	VW	0.30 [3.71]	0.35 [4.33]	0.29 [3.61]	0.29 [3.53]	0.28 [3.47]	0.28 [3.45]
Quintile	Cap	VW	0.20 [2.21]	0.27 [3.04]	0.21 [2.44]	0.21 [2.32]	0.19 [2.20]	0.19 [2.17]
Decile	NYSE	VW	0.25 [2.56]	0.30 [3.14]	0.21 [2.26]	0.20 [2.14]	0.24 [2.60]	0.24 [2.50]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.23 [2.78]	0.29 [3.60]	0.23 [2.89]	0.23 [2.95]	0.21 [2.64]	0.22 [2.79]
Quintile	NYSE	EW	0.29 [3.44]	0.40 [5.02]	0.30 [4.34]	0.27 [3.88]	0.15 [2.27]	0.14 [2.09]
Quintile	Name	VW	0.26 [3.22]	0.31 [3.78]	0.24 [3.11]	0.24 [3.08]	0.24 [3.01]	0.24 [3.07]
Quintile	Cap	VW	0.16 [1.80]	0.23 [2.59]	0.17 [2.00]	0.17 [1.95]	0.16 [1.92]	0.16 [1.92]
Decile	NYSE	VW	0.21 [2.13]	0.26 [2.62]	0.17 [1.85]	0.17 [1.79]	0.20 [2.12]	0.20 [2.16]

Table 3: Conditional sort on size and EIR

This table presents results for conditional double sorts on size and EIR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EIR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EIR and short stocks with low EIR. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EIR Quintiles					EIR Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.44 [1.60]	0.74 [2.77]	0.86 [3.21]	0.94 [3.72]	1.11 [4.56]	0.66 [5.72]	0.74 [6.44]	0.64 [6.02]	0.61 [5.66]	0.46 [4.39]	0.45 [4.24]
	(2)	0.55 [2.21]	0.74 [2.98]	0.79 [3.28]	1.01 [4.27]	1.00 [4.41]	0.43 [4.15]	0.51 [5.09]	0.39 [4.21]	0.36 [3.80]	0.32 [3.40]	0.30 [3.14]
	(3)	0.67 [2.92]	0.67 [2.87]	0.84 [3.67]	0.94 [4.27]	0.93 [4.52]	0.26 [2.89]	0.34 [3.79]	0.24 [2.86]	0.23 [2.66]	0.18 [2.08]	0.17 [1.99]
	(4)	0.49 [2.25]	0.68 [3.15]	0.85 [4.02]	0.88 [4.36]	0.89 [4.63]	0.41 [4.18]	0.49 [5.15]	0.36 [4.34]	0.37 [4.38]	0.22 [2.67]	0.24 [2.90]
	(5)	0.55 [3.09]	0.49 [2.68]	0.44 [2.44]	0.57 [3.36]	0.77 [4.77]	0.20 [2.19]	0.25 [2.69]	0.20 [2.16]	0.21 [2.25]	0.24 [2.55]	0.25 [2.63]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EIR Quintiles					EIR Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	315	314	314	313	313	25	28	32	24	24	
	(2)	91	90	90	90	89	48	49	50	49	48	
	(3)	65	65	65	64	65	85	85	86	88	88	
	(4)	55	55	55	55	55	185	187	196	194	199	
(5)	51	51	51	51	50	1352	1347	1644	1482	1780		

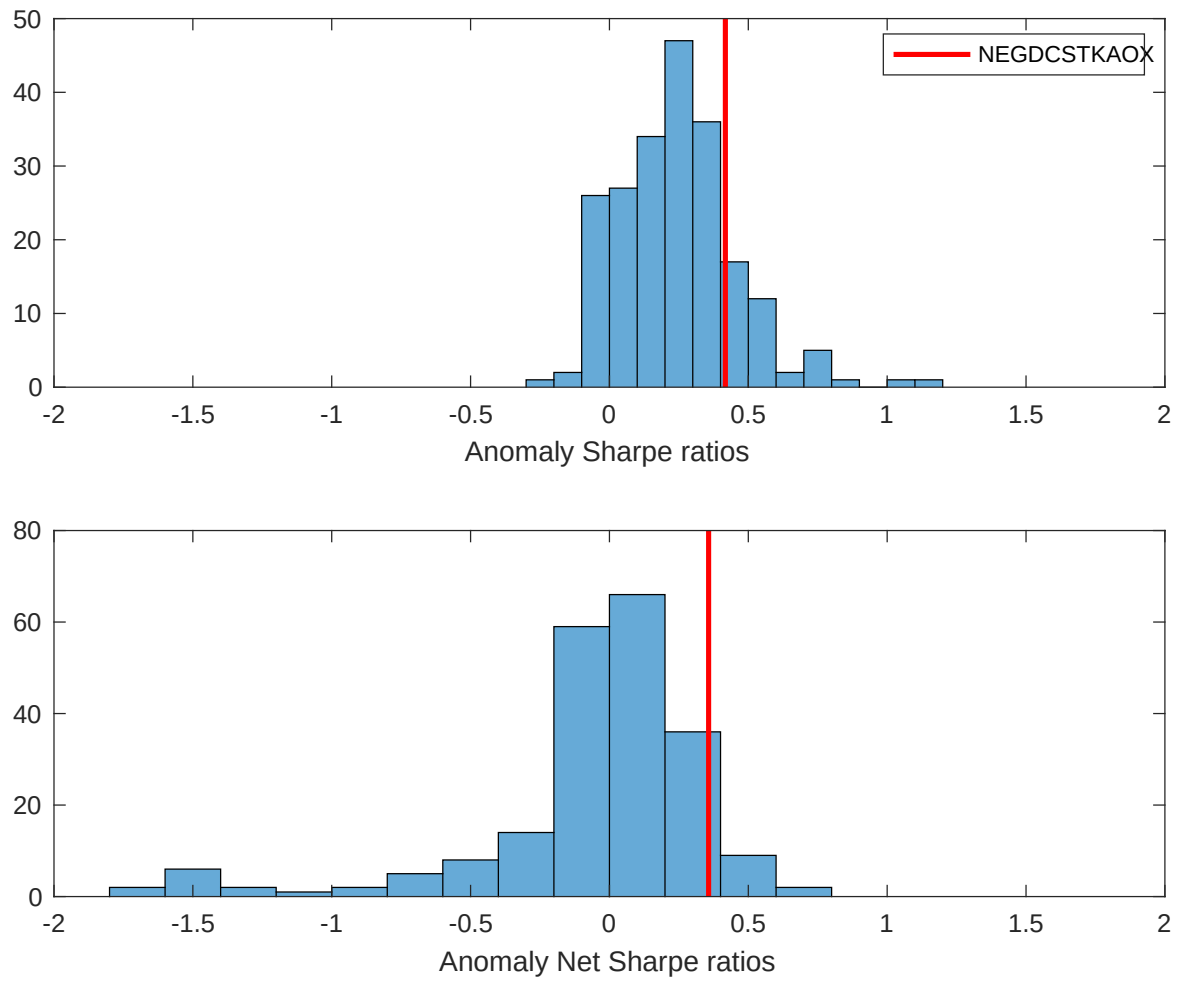


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EIR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

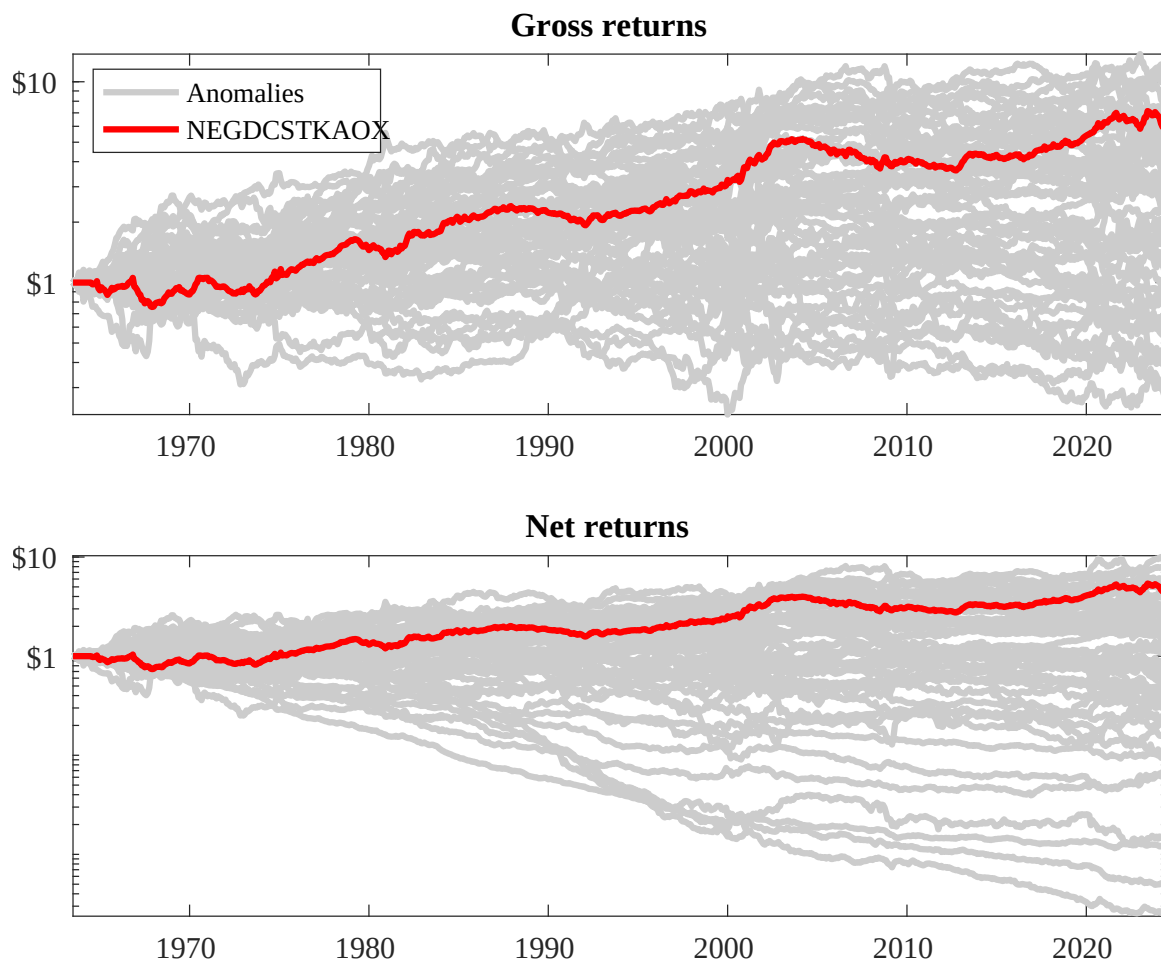
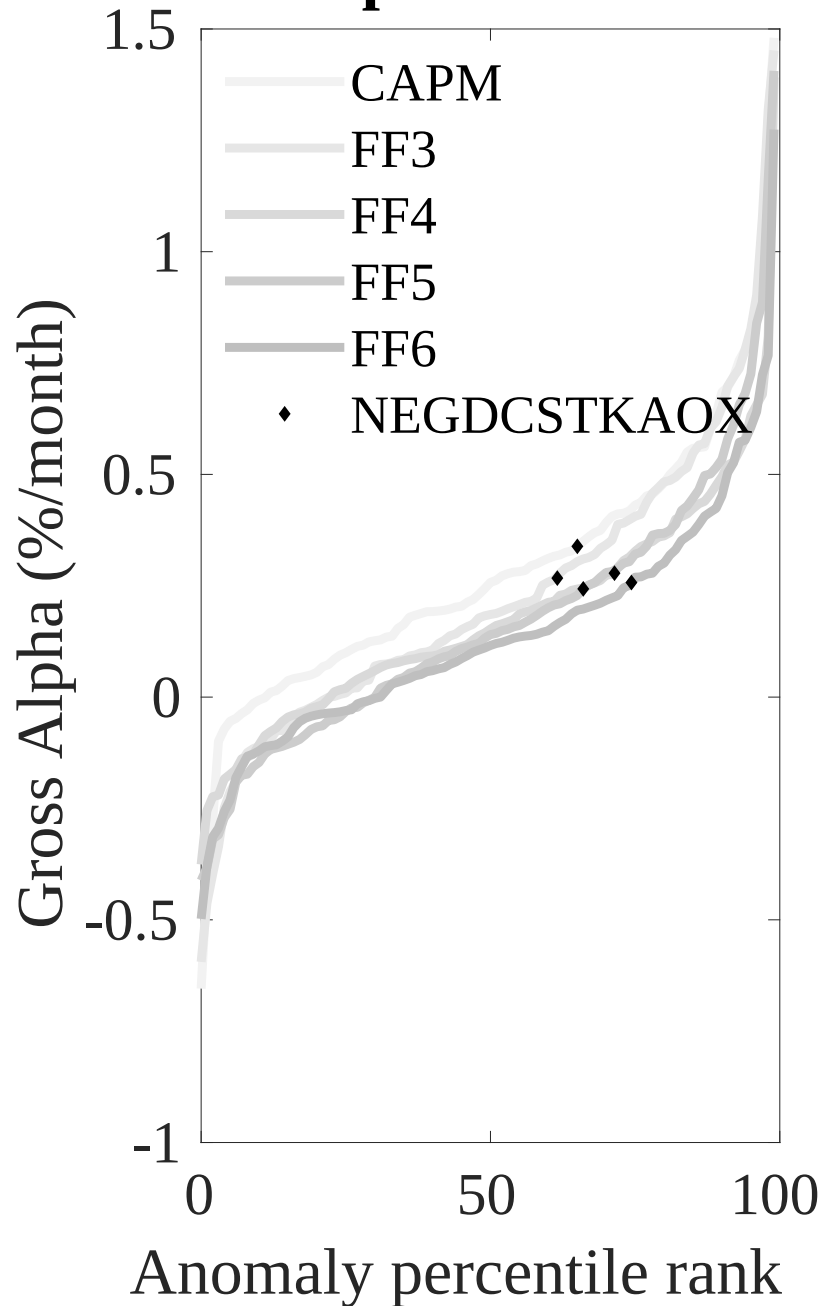


Figure 3: Dollar invested.

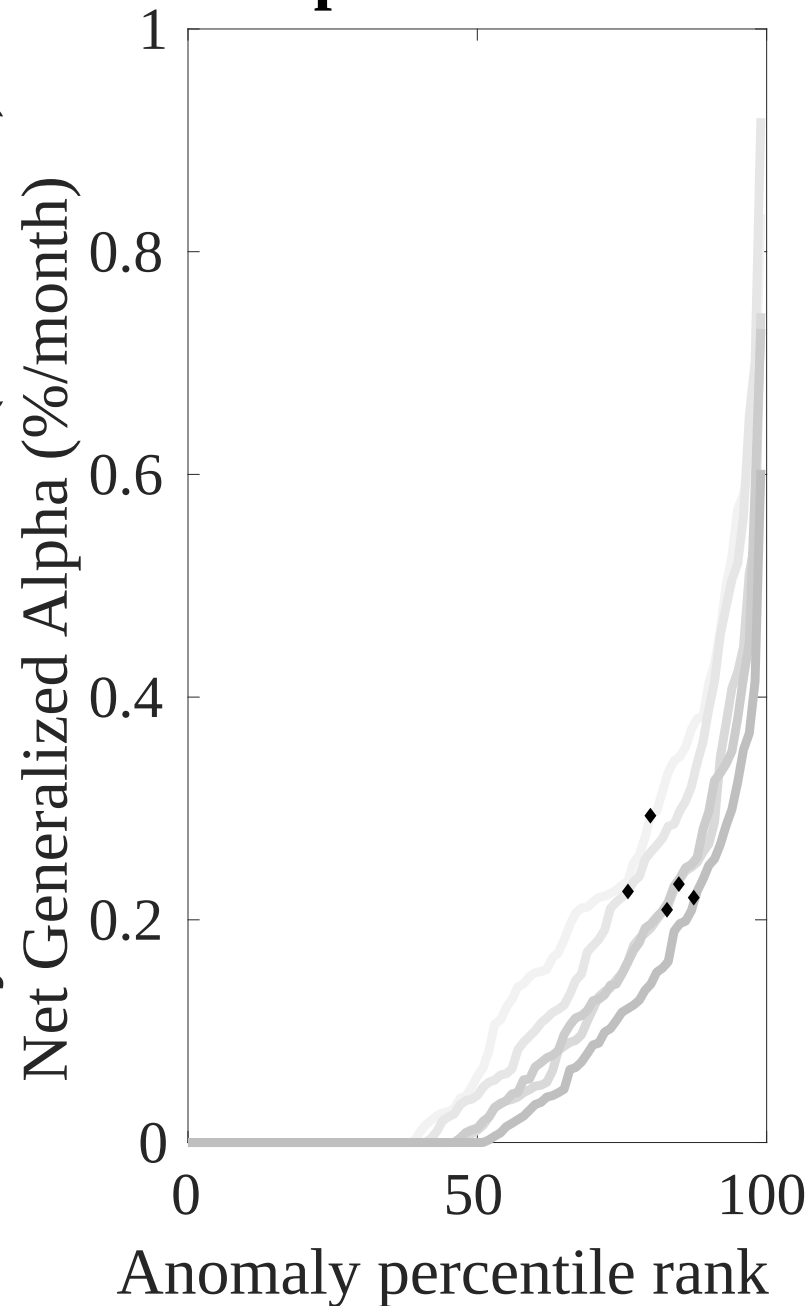
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EIR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



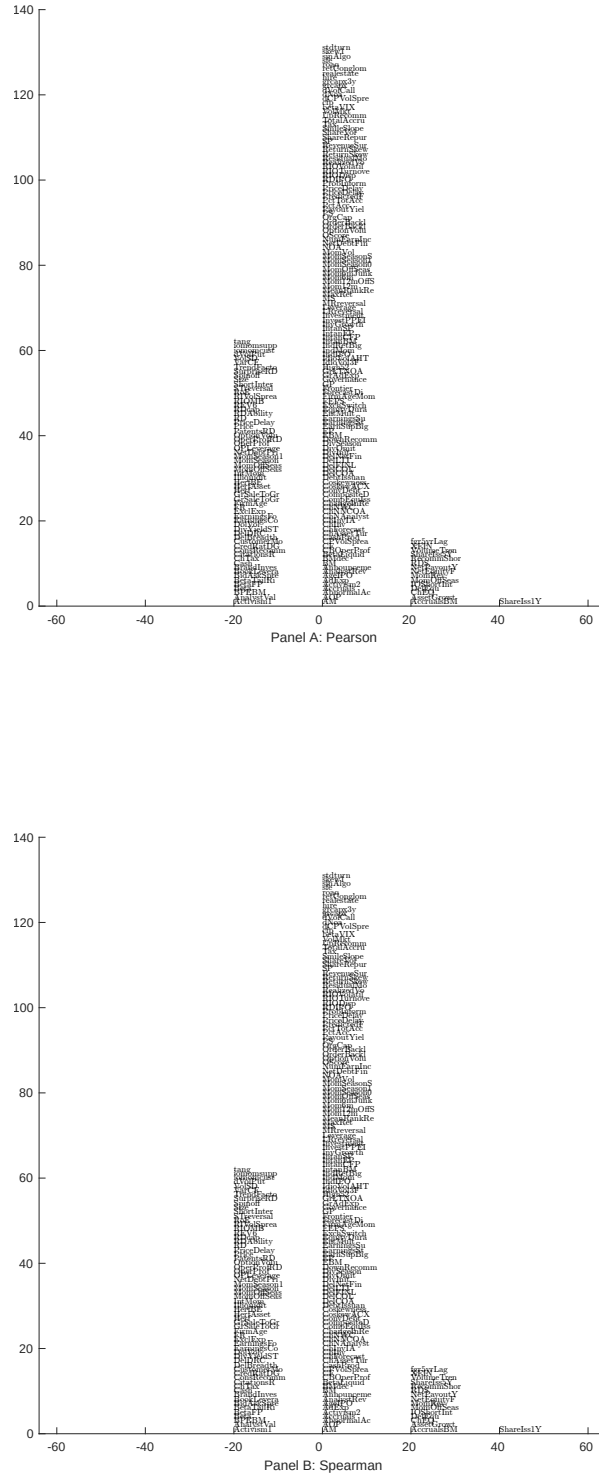


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 209 filtered anomaly signals with EIR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

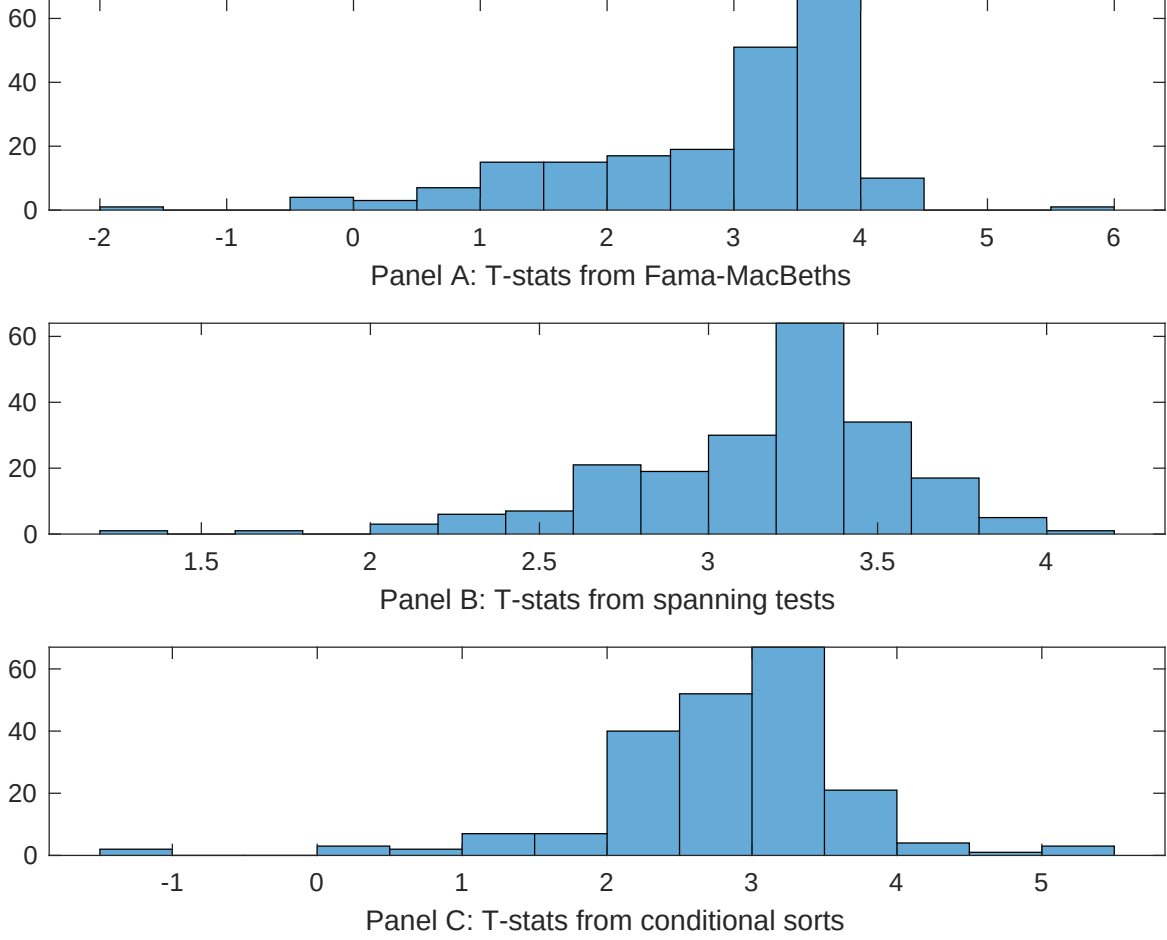


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EIR conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EIR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EIR}EIR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EIR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EIR. Stocks are finally grouped into five EIR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EIR trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EIR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EIR}EIR_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Net Payout Yield, Growth in book equity, Share issuance (1 year), Change in equity to assets, Share issuance (5 year), Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.12 [5.34]	0.18 [7.25]	0.13 [5.84]	0.13 [5.54]	0.14 [6.23]	0.13 [5.31]	0.12 [4.61]
EIR	0.30 [1.29]	0.46 [2.04]	0.57 [2.72]	0.63 [2.71]	0.56 [2.81]	0.53 [2.21]	0.30 [1.22]
Anomaly 1	0.24 [2.30]						0.14 [2.78]
Anomaly 2		0.52 [5.12]					-0.16 [-1.12]
Anomaly 3			0.16 [6.93]				0.90 [3.85]
Anomaly 4				0.16 [4.76]			0.13 [2.18]
Anomaly 5					0.29 [5.40]		0.28 [6.46]
Anomaly 6						0.15 [2.40]	-0.15 [-1.83]
# months	703	708	703	708	703	630	622
$\bar{R}^2(\%)$	1	0	0	0	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EIR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EIR} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Net Payout Yield, Growth in book equity, Share issuance (1 year), Change in equity to assets, Share issuance (5 year), Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.28 [3.60]	0.23 [3.01]	0.24 [3.04]	0.25 [3.15]	0.19 [2.41]	0.29 [3.47]	0.26 [3.23]
Anomaly 1	20.23 [6.65]						17.33 [4.24]
Anomaly 2		32.24 [7.59]					32.13 [4.93]
Anomaly 3			17.48 [4.32]				-9.66 [-1.81]
Anomaly 4				17.24 [4.32]			-6.74 [-1.12]
Anomaly 5					23.63 [5.57]		17.10 [3.23]
Anomaly 6						12.43 [3.24]	-6.78 [-1.41]
mkt	-3.12 [-1.63]	-5.19 [-2.81]	-5.75 [-3.03]	-6.55 [-3.45]	-5.13 [-2.72]	-4.79 [-2.30]	-3.32 [-1.63]
smb	-0.92 [-0.34]	-5.26 [-1.97]	-3.66 [-1.33]	-4.40 [-1.61]	-2.73 [-1.00]	3.09 [0.93]	-1.79 [-0.55]
hml	4.55 [1.19]	8.64 [2.39]	12.26 [3.34]	10.31 [2.78]	14.54 [3.99]	11.48 [3.02]	5.38 [1.39]
rmw	-16.52 [-4.07]	-2.90 [-0.80]	-14.08 [-3.27]	-2.90 [-0.78]	-13.13 [-3.30]	-8.43 [-1.87]	-7.88 [-1.61]
cma	4.49 [0.77]	-10.20 [-1.52]	10.38 [1.75]	3.82 [0.57]	9.87 [1.73]	7.41 [1.20]	-21.25 [-2.93]
umd	-0.36 [-0.19]	-2.46 [-1.34]	-2.70 [-1.43]	-1.48 [-0.79]	-3.45 [-1.83]	-1.98 [-1.01]	-1.92 [-1.01]
# months	716	720	716	720	716	630	626
$\bar{R}^2(\%)$	21	22	18	18	19	14	23

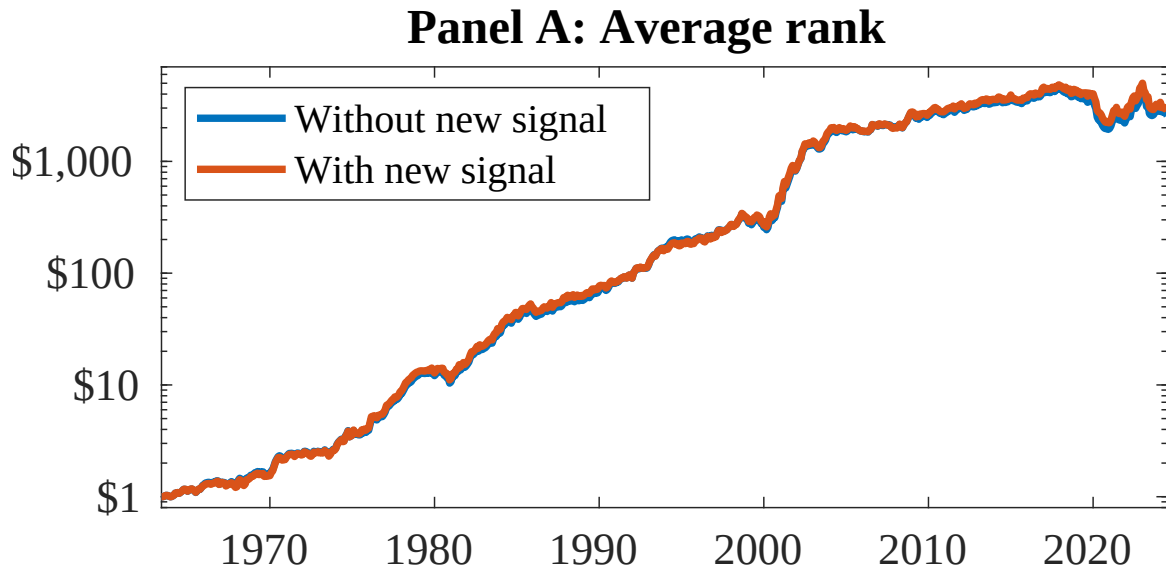


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EIR. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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